



JULIO LANDA

Mechatronics engineer
Master's degree in renewable energy

CONTACT

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LANGUAGES

- Spanish – Native
- English – Upper-intermediate/advanced

SKILLS

- **Instrumentation:** Design and calibration of instruments.
- **Maintenance:** Preventive and corrective maintenance.
- **Electrical panels:** Design and assembly of electrical systems
- **Programming:** Python, C/C++, Arduino, LabVIEW.
- **Automation:** Design and development of industrial control systems.
- **Mechanical design:** CAD modeling with SolidWorks and FreeCAD.

PROFESSIONAL EXPERIENCE

NC SCHOOL CUERNAVACA 2024 - 2025

Science Lab Coordinator & Teacher.

- Lead teacher of Physics and Mathematics.

IER, UNAM

Teaching assistant. 2023 - 2025

- Assisted in grading assignments, exams, and projects for the course *Introduction to Bioclimatic Design* (Renewable Energy Engineering, 6th semester).

IoT developer engineer. 2020 / 2024 - present

- Developed low-cost devices with open-source hardware and software, integrated with IoT platforms for the instrumentation project of the new Renewable Energy Engineering building at the IER, UNAM.

Workshop coordinator at CNEER. 2023

- Planned, organized, and coordinated workshops for the 10th edition of CNEER (Congreso Nacional de Estudiantes de Energías Renovables).

DMC INGENIERÍA 2022

Electrical engineer.

- Provided maintenance to electric motors and control panels as a contractor for Cementos Moctezuma.

BOMBAS Y SERVICIOS CORBALA 2021

Electrical engineer.

- Trabajé dando mantenimiento e instalando bombas de agua para pozos profundos, así como diseñando e instalando sistemas y tableros de control.

ALUCAPS 2020

Maintenance technician.

- Applied maintenance plans to company machinery, updated and programmed production line control systems using PLCs, and assembled electrical panels.

- **Data analysis:** Procesing and visualization with Python (Numpy, Pandas, SciPy, Sk-Learn, Matplotlib, Seaborn, Plotly).
- **Artificial intelligence:** Machine learning, neural networks, genetic algorithms.
- **IoT (Internet of Things):** Development with ESP32/Arduino and integration with IoT platforms.
- **Electronics:** Circuit design and automation projects.
- **PLC's:** Programming of control systems.

EDUCATION

MASTER'S IN ENGINEERING – ENERGY AREA 2022 - 2025
Instituto de Energías Renovables de la UNAM, Morelos, México.

BACHELOR OF MECHATRONICS ENGINEERING 2018 - 2020
Universidad Tecnológica de Emiliano Zapata, Emiliano Zapata, Morelos México.

HIGHER UNIVERSITY TECHINICIAN IN MECHATRONICS. 2016 - 2018
Universidad Tecnológica de Emiliano Zapata, Emiliano Zapata, Morelos México.

PUBLICATIONS

ARTICLE - IOT SMARTWATCH BASED ON OPEN TECHNOLOGIES FOR THE COLLECTION OF THERMAL COMFORT DATA
HardwareX. 2025
DOI: <https://doi.org/10.1016/j.ohx.2025.e00633>

THESIS - IOT SMARTWATCH BASED ON OPEN TECHNOLOGIES FOR THE COLLECTION OF THERMAL COMFORT DATA
Master's thesis. 2025
Available at:
https://altamarmx.github.io/reloj_inteligente/
Project licensed under GPL v3.0:
https://github.com/JulioLanda4/IoT_Reloy_Confort/tree/v1.0.0

CERTIFICATIONS, COURSES, HONORS & AWARDS

CERTIFICATIONS

- *Certified Solidworks Professional (CSWP).* 2019
- *Google IT Automation with Python.* 2022
- *LabVIEW Associate Developer.* 2019

COURSES

- *Macrotraining in Artifical Intelligenece (MeIA).* - 2023
- *Python for engineering (UNAM, Coursera).* - 2023

HONORS & AWARDS

- *1st place – 3rd Mathematics Olympiad, Universidad Tecnológica de Emiliano Zapata. - 2019*
- *Honorable Mention – XXXIII CUAM-ACMor Research Congress (Physics-Mathematics category), project Ecological Greenhouse from Plastic Waste - 2023.*
- *Recognition as speaker at the Sustainable Buildings Symposium: Simulations and Digitalization for Decarbonization - 2024.*
- *Honorable Mention – XXXV CUAM-ACMor Research Congress (Physics-Mathematics category), project Green Motion: How Do You Move? - 2025.*